

Concurrency and Distributed Systems

Week 8 – Part 3: Performance Tradeoff and Technical Explanation

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Part 3 Roadmap

- 1 Tradeoff
- 2 Recorded Explanation
- 3 Demo

Every Fix Has Cost

Benefit

Timeout bounds waiting and protects request threads.

Cost

A slow real response may be replaced by a fallback response.

Measure

Latency percentiles, timeout count, fallback rate, and error rate.

Measure Before Optimizing

- Do not claim faster without measurement.
- Separate throughput, latency, and correctness.
- Watch tail latency, not only averages.
- Count fallbacks as a degraded-service signal.

Five-Minute Structure

- ① Bug: what broke.
- ② Evidence: what proved it.
- ③ Mechanism: why it happened.
- ④ Fix: what changed.
- ⑤ Cost: what the fix trades away.

Rubric

Area	Points
Bug	25
Evidence	20
Mechanism	25
Fix	15
Cost	15

Run

```
.\run-demo.ps1 04
```

- Identify benefit.
- Identify cost.
- Identify measurement plan.